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Studierichting: Informatica
Oefeningenreeks 1B nr. 9.5

BVW:

$x + 4 > 0 \Rightarrow x > -4$

$x - 2 > 0 \Rightarrow x > 2$

$x > 0$

$\log_3(x+4) + \log_3(x-2) = 2\log_3(x)$

$\Leftrightarrow \log_3(x+4) + \log_3(x-2) + \log_3\left(x^{-2}\right) = 0$

$\Leftrightarrow \log_3\left(\frac{(x+4)(x-2)}{x^2}\right) = 0$

$\Leftrightarrow \log_3\left(\frac{x^2 - 2x + 4x - 8}{x^2}\right) = 0$

$\Leftrightarrow \frac{x^2 + 2x - 8}{x^2} = 1$

$\Leftrightarrow x^2 + 2x - 8 = x^2$

$\Leftrightarrow 2x - 8 = 0$

$\Leftrightarrow x = 4$

References